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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
    - Profile setup: Create a user account (e.g., server / server123)
    - Install OpenSSH server: **Check this option** (Important!)
    - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode
4. Click **OK**

Install/Verify SSH Server

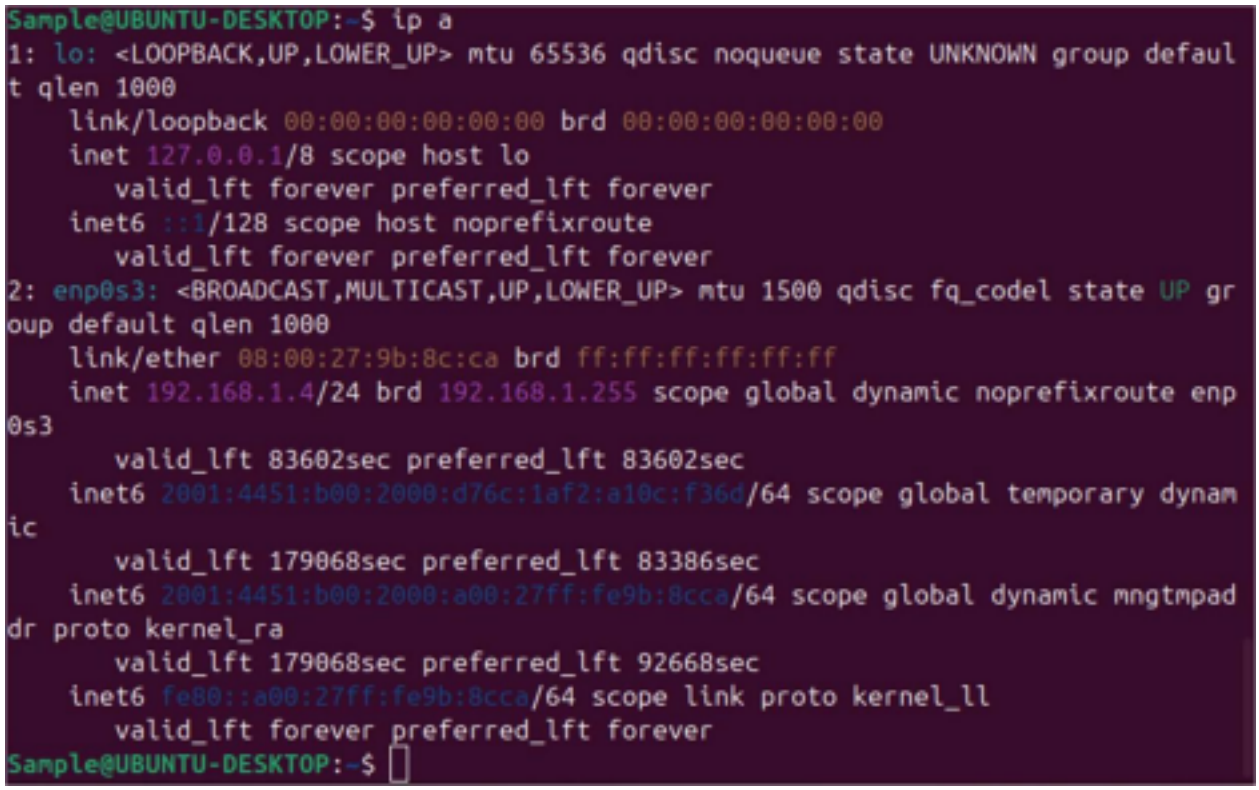
Server: Install SSH on the server (if not already installed):

- sudo systemctl status ssh – for checking the SSH status.
- sudo apt update – updating the your server.
- sudo apt install openssh-server -y –installing the SHH.
- sudo systemctl enable ssh – enabling the SSH to your unit. (SERVER/DESKTOP)
- sudo systemctlstartssh – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

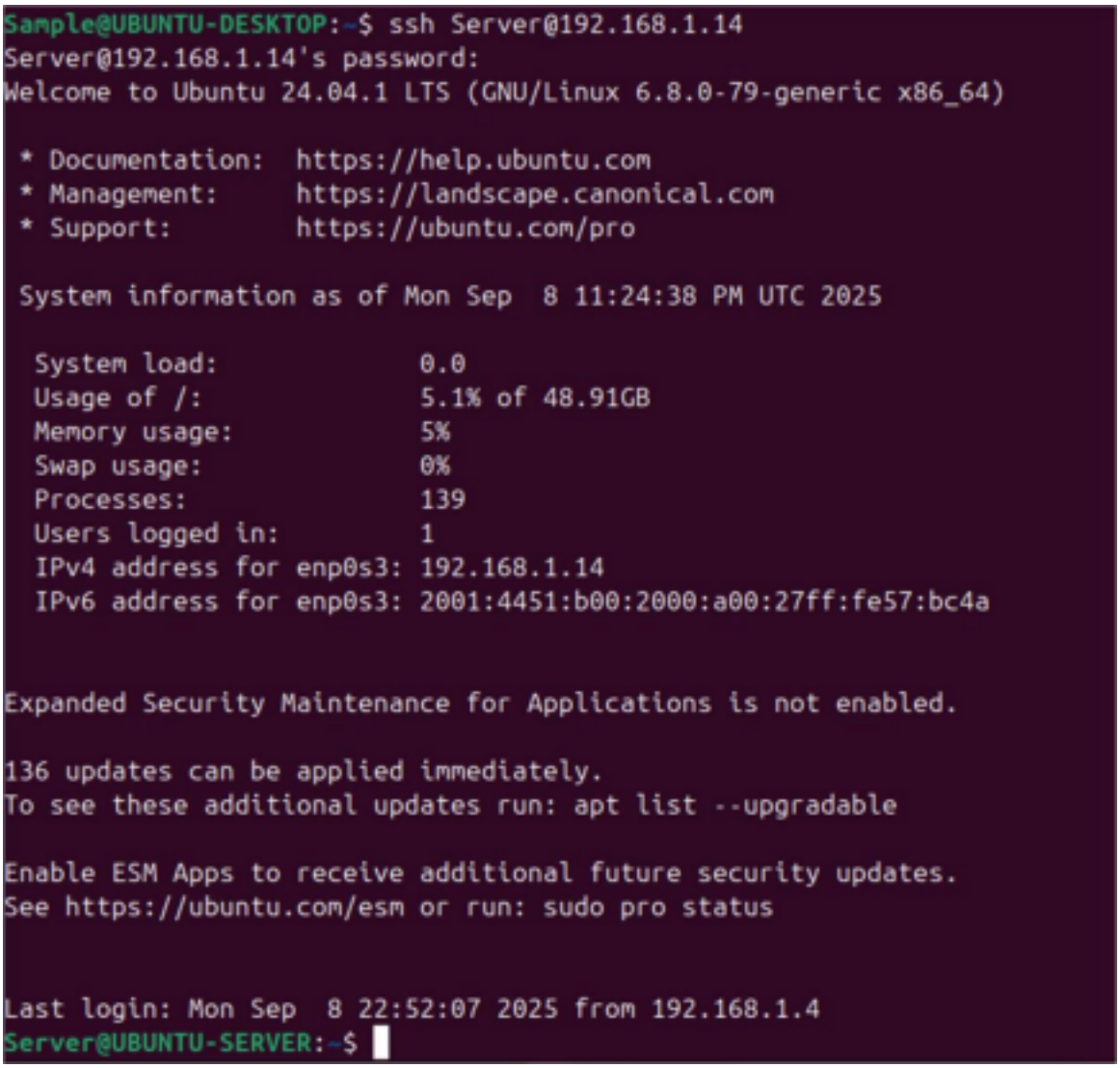
Inside the VM, run:



From the desktop, connect to the server: ssh username@192.168.1.120

2. Connect from Desktop

From your host desktop:



Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of ip a or hostname -I)
- SSH service status (sudo systemctl status ssh)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Date Submitted: 09/02/26
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1 OBJECTIVES (5 POINTS)

Check all that apply:

□w Create and configure a Virtual Machine with Ubuntu Server
□ Configure Bridged Network Adapter for network access
□ Determine and verify the server's IP address
□ Install OpenSSH Server and enable the service
□ Establish successful SSH connection from host machine
□ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu
Virtualization Software (VirtualBox/VMware)	Oracle Virtual Box
Ubuntu Server Version	Ubuntu-26.04 live server
VM Name	uncleserver
RAM Allocated	8000
Storage Allocated	100

Network Mode NAT Network

SSH Client Used Secure Shell Client

3 VM CONFIGURATION

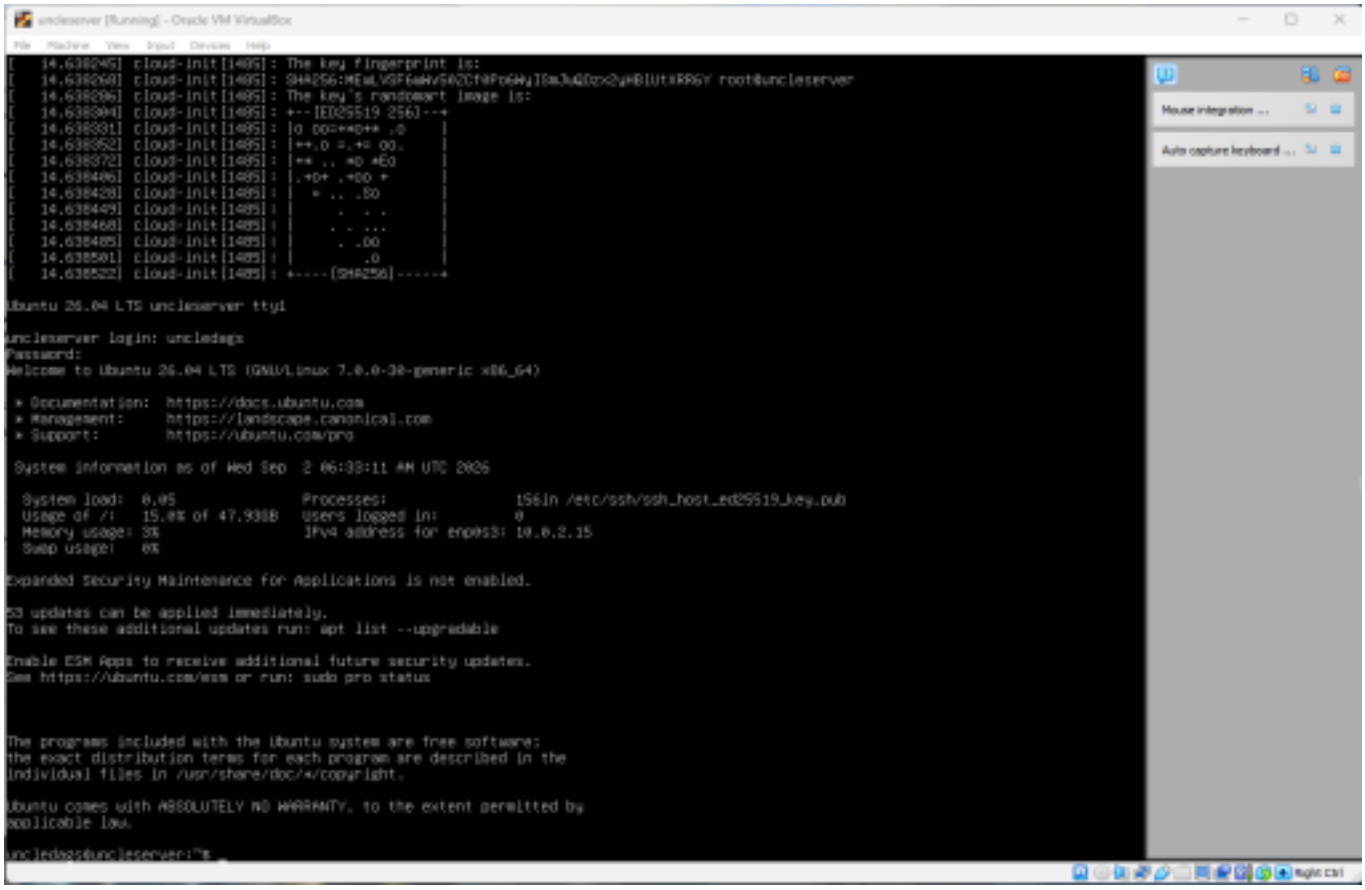
Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	8000 MB
Hard Disk	100 GB
Network Adapter	Bridged

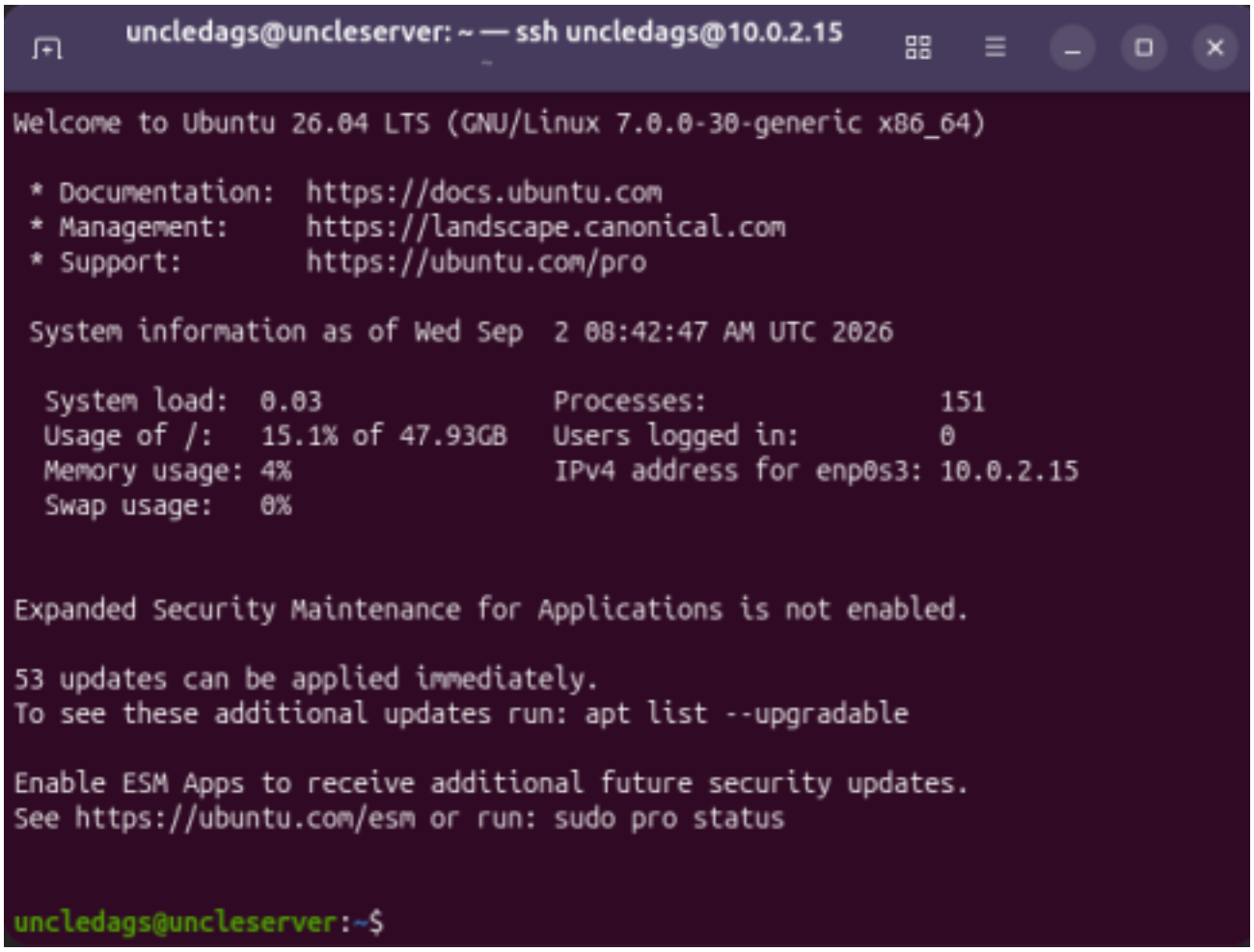
4. NETWORK CONFIGURATION
Commands Executed:
Server IP Address: 10.0.2.4
Network Interface: NAT Network
6. REMOTE CONNECTION
Connection Details:

Parameter	Username
Server IP	SSH Port

Value

10.0.2.15 uncledags 22




uncledags@uncleserver: ~ — ssh uncledags@10.0.2.15
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Wed Sep 2 08:42:47 AM UTC 2026
System load: 0.03
Usage of /: 15.1% of 47.93GB
Memory usage: 4%
Swap usage: 0%
Processes: 151
Users logged in: 0
IPv4 address for enp0s3: 10.0.2.15
Expanded Security Maintenance for Applications is not enabled.
53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
uncledags@uncleserver:~$